

MIS-Motorcycle



Welcome and thank you for supporting MIS-Motorcycle!

Contents

1. [Quick Start](#)
2. [Setup a character](#)
3. [Setup a motorcycle](#)
4. [Upgrade](#)
5. [Contacts Information](#)

1. Quick Start

1. API Compatibility Level

Edit – Project Settings – Player – Other Settings

■ Unity 2019/2020

Api Compatibility Level* .NET 4.x ▼

■ Unity 2021

Api Compatibility Level* .NET Framework ▼

2. Setup Sequence

- Post-Process (Optional)
- INVECTOR
- MIS
- MIS-Motorcycle

3. Demo Scene

Find Assets/MIS-Packages/Motorcycle/Demo/.

 [21:36:14] Vehicle layer has been added on the 30th layer mask index.

NOTE: When this package is imported, Vehicle layer is automatically registered on the 30th index.

4. Play

Control	Input
Get On / Off	E / Q
Move	WASD
Brake (Instant)	S (While moves forward) W (While moves backward)
Hand Brake (Drift)	Holding down Space
Boost	Holding down Left SHIFT
Jump	Left CTRL
Weapon Mode Change (Only for Shooter Template)	CapsLock

NOTE: Every input key can be modified via Inspector.

2. Setup a Character

1. Overall

MIS Package Setup usually proceeds as following tasks when this package needs:

- Replaces Invector components with MIS's.

If MIS controller and input manager are already installed, this step will be skipped.

- Adds Animator Parameters

If MIS Animator Parameters are already added, this step will be skipped.

- Generates Animator StateMachine / States

If the relevant Animator Layers/StateMachines/States are already added, each step will be skipped.

Any Animator layouts of this package setup are automatically arranged.

- Adds MIS Package components and extra settings such as VFX.

If the relevant components or extra settings are already installed, this step will be skipped.

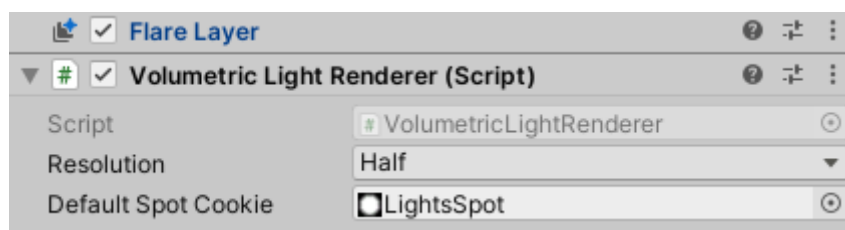
- Sometimes MIS Setup modifies components default value to fix any problem. Please see the console log.

You can run this setup repeat on the same character although the character already has been set up.

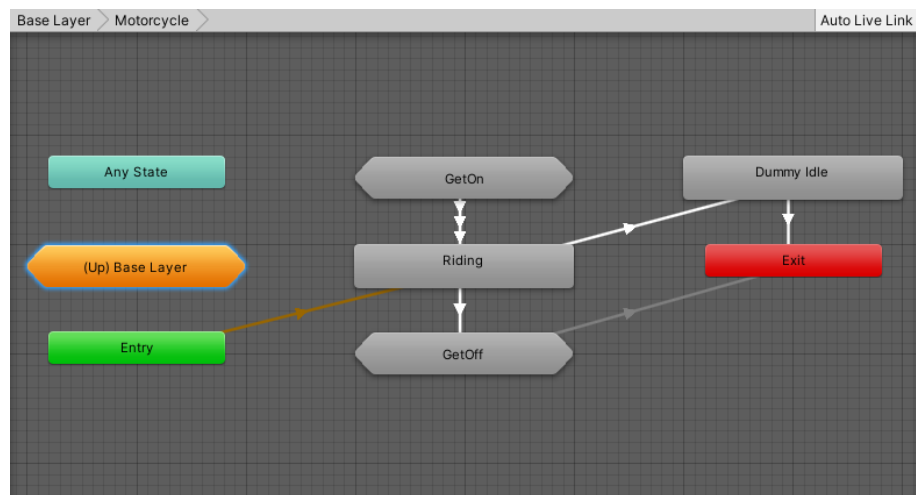
2. mvMotorcycleRiderShooter



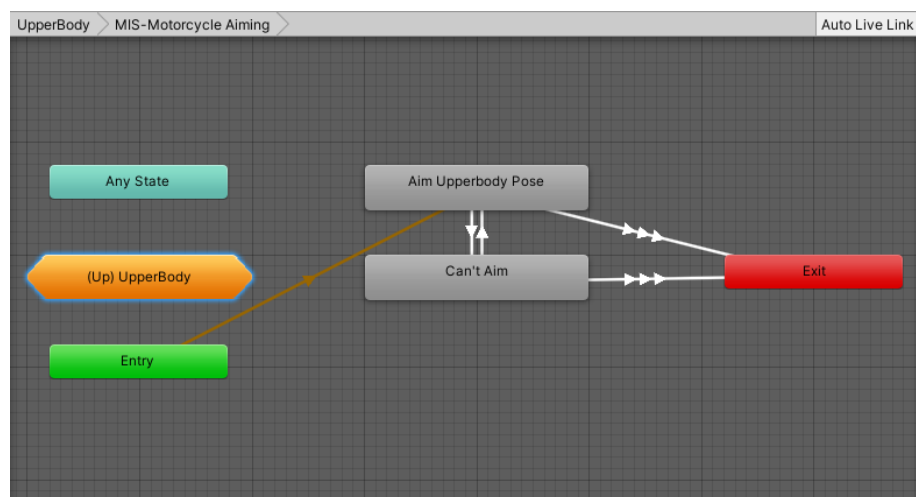
■ Main Camera



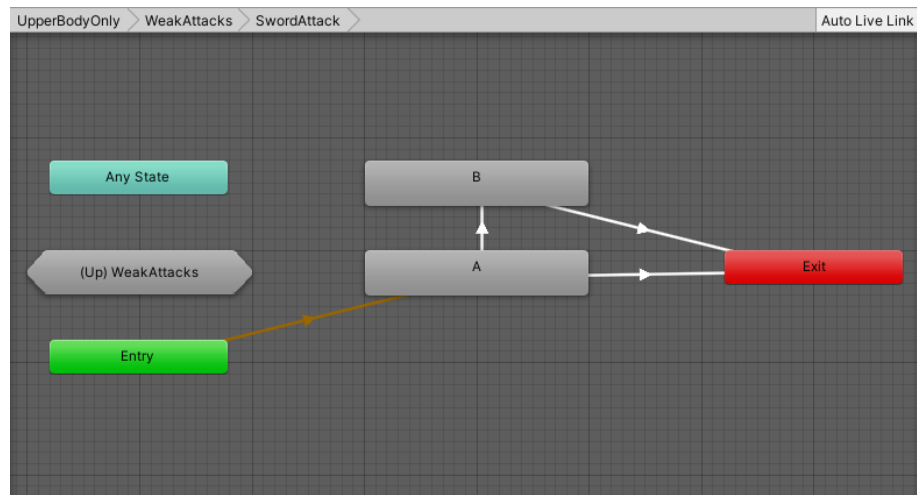
3. Animator



<Basic/Melee/Shooter>



<Melee/Shooter>



<Melee/Shooter>

3. Setup a Motorcycle

1. MIS-Motorcycle Basic Setup

MIS-Motorcycle works based on WheelCollider. Rigidbody complements the lack of WheelCollider.

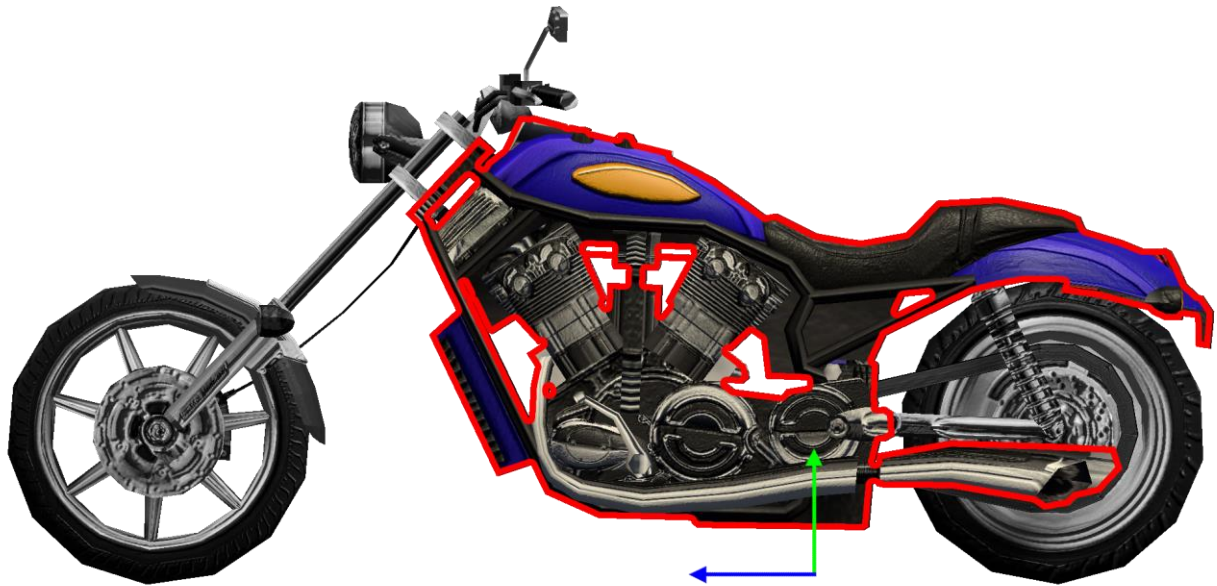
MIS-Motorcycle supports Anti-Understeering, Anti-Oversteering, Anti-Overturn, and ESP internally, but it is not as important as the following basic setup.

Mesh, its pivot position & direction, and Center of Mass of Rigidbody are very important because it basically works with physics. Even after setup is complete, further adjustments may be required to achieve the desired behavior. You must check the Harley motorcycle mesh provided in this package.



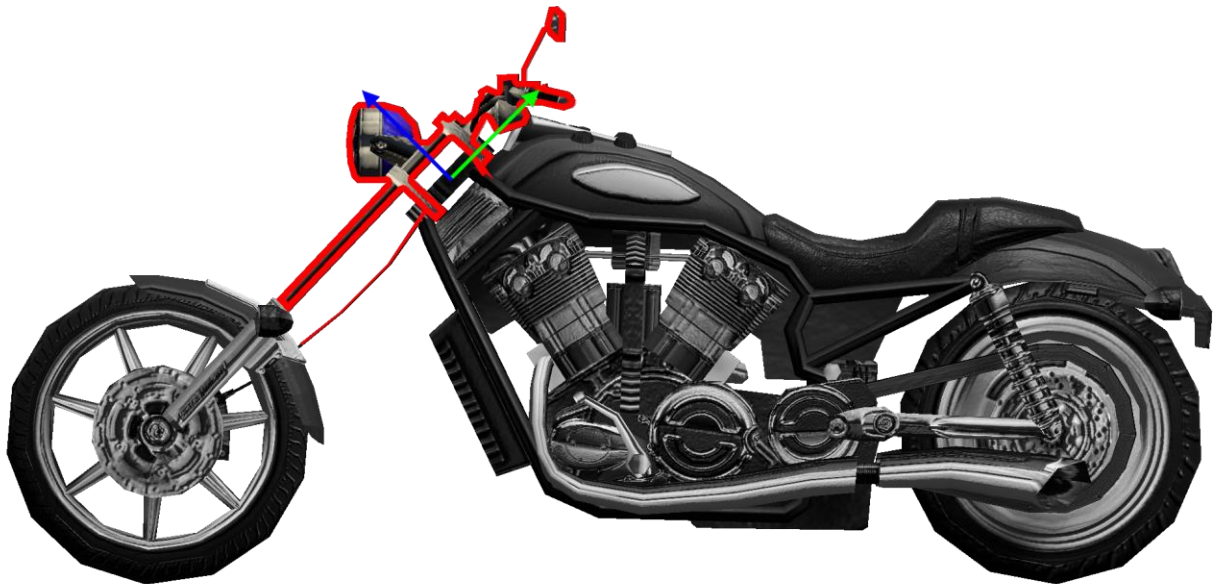
<Center of Mass>

It takes a very important role in motorcycle's behavior in various situations such as move, turn, jump, collision, and etc.



<Body>

It is responsible for rolling according to the steer angle.



<Steer>

The pivot must be set for the steer position, rotation and direction.



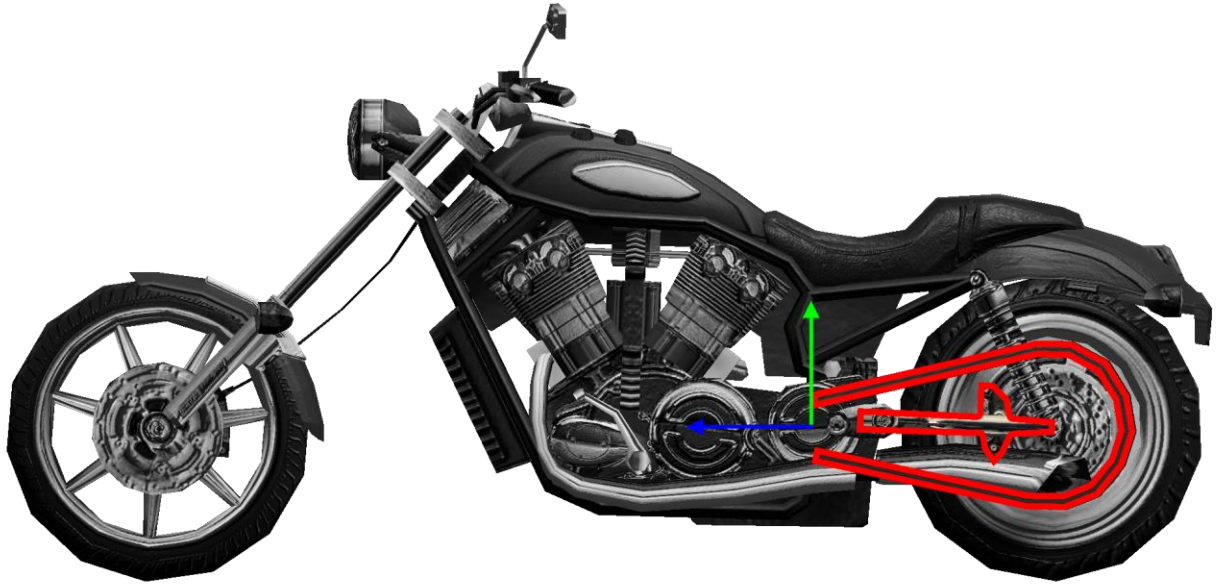
<Front Suspension>

It is responsible for the front wheel movement according to WheelCollider suspension motion.

The pivot position and direction are also very important for the movement.



<Front Wheel>



<Rear Axle>

It is responsible for the rear wheel movement according to WheelCollider suspension motion.

The pivot position and direction are also very important for the movement.



<Rear Suspension>

It is responsible for the rear wheel movement according to WheelCollider suspension motion.

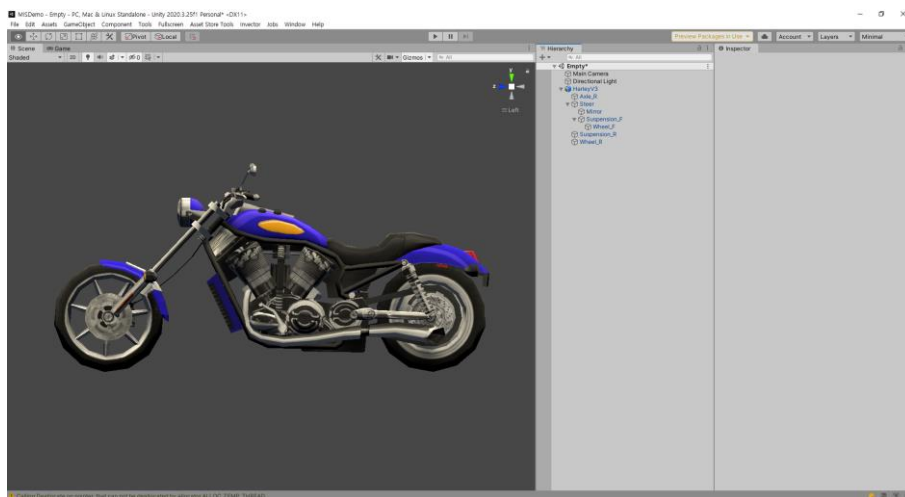
The pivot position and direction are also very important for the movement.



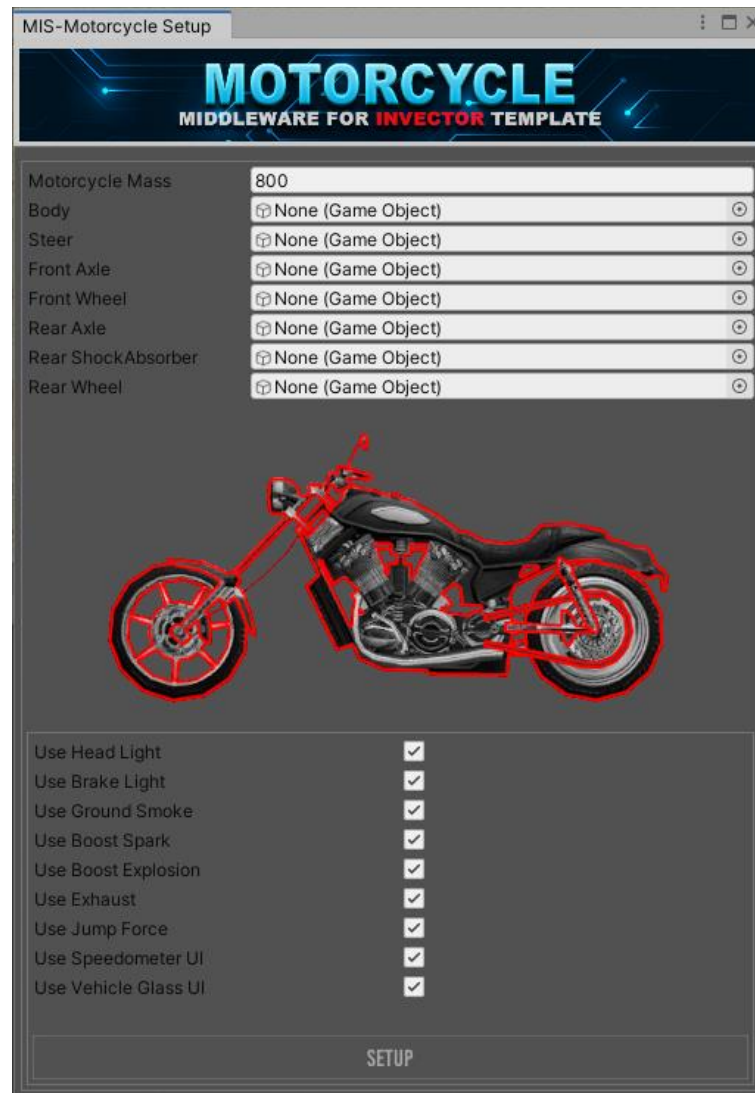
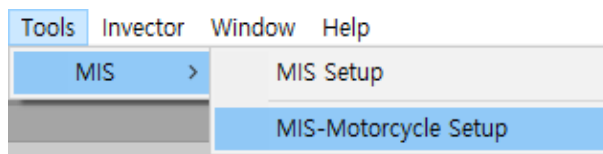
<Rear Wheel>

2. Basic MIS-Motorcycle Setup

Place your motorcycle mesh in Hierarchy.



3. Motorcycle Setup Window



Register each part to the corresponding slot.

Axle Front, Rear Axle, and Rear Shock Absorber parts are optional. The parts are animated according to WheelColliders' suspension movement.

4. Motorcycle Setup

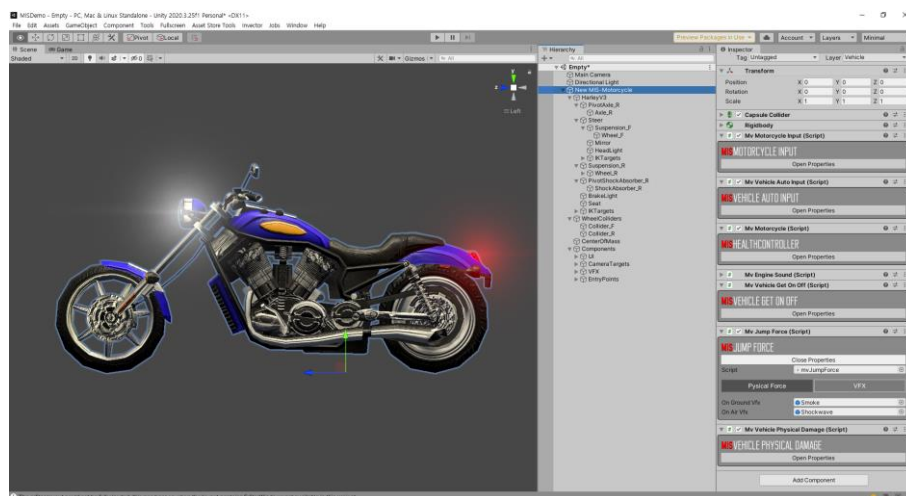
Click the SETUP button.

This setup automatically organizes the parts configuration. It also installs various components and effects required for each part. Other parts not used in this setup remain under the Body object.

IMPORTANT:

Unlike Character Setup, Motorcycle Setup cannot be executed repeatedly on the current setup that's because all parts are reorganized. Please use a new motorcycle mesh each time.

5. Motorcycle Basic Setup completed



All parts are newly organized and all necessary components are installed.

6. mvMotorcycle

#

✓

Mv Motorcycle (Script)

?

⌵

⋮

MISHEALTHCONTROLLER

Close Properties

Script

mvMotorcycle

Health

Settings

Vehicle

Core

Events

Debug

Stamina

Max Stamina

Origin

500

Currer

500

Stamina Recovery

Origin

1.2

Currer

1.2

Inventory Camera Position

Inventory Camera Position

-0.15

Jump

Max Jump Count

Origin

3

Currer

3

Jump Stamina

20

Jump Stamina Recovery D

2

Jump Duration

Origin

0.4

Currer

0.4

Jump Height

5

Jump Force

8

Boost

Boost Multiplier

2

Boost Stamina

15

Bost Stamina Recovery De

2

Jump Boost Force Multiplie

5

Overtuned

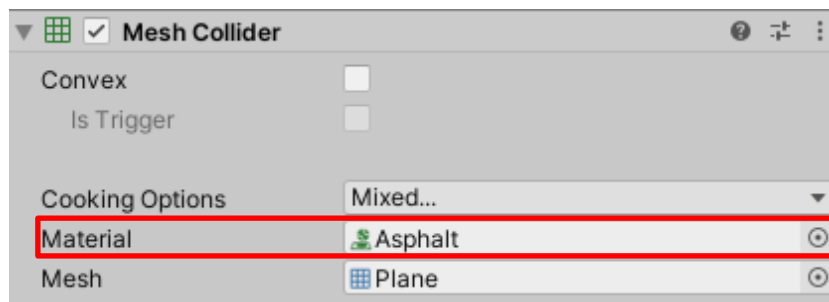
Overtuned Angle

55

7. Details Setup

8. Physics Materials for Ground Frictions

Find Assets/MIS-Packages/Vehicle/Runtime/VFX/Physics Materials / folder, and then set it into the Physics Material slot on the ground object. Various effects such as skidmarks and smoke are applied differently depending on the ground material.

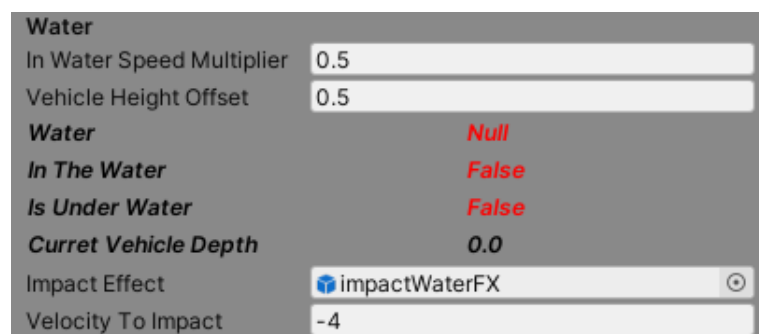


9. FlareLayer

Even if the Volumetric Headlight has not been applied, it is recommended to apply FlareLayer component to the main camera. It makes the head and brake lights look more realistic.



10. INVECTOR Swimming addon



11. Vehicle Height Offset

When this position goes below the water level, it becomes 'Is Under Water'.



12. Impact Effect

The water impact effect will be assigned automatically if the INVECTOR Swimming feature is enabled.



13. At last, detailed settings of each installed components are required. Please watch [MIS-Motorcycle Setup Guide](#) video to set each of the rest items one by one.

14. Physics – Layer Collision Matrix

▼ Layer Collision Matrix

	PostProcess	Vehicle	CoverPoint	Pushable	BlockAIRayCast	BodyPart	HeadTrack	StopMove	Trippers	CompanionAI	Enemy	Player	UI	Water	Ignore Raycast	TransparentFX	Default
Default	✓	✓		✓	✓	✓			✓	✓	✓	✓		✓	✓	✓	✓
TransparentFX	✓	✓			✓	✓				✓	✓	✓				✓	
Ignore Raycast	✓	✓			✓	✓				✓	✓	✓			✓		
Water	✓	✓			✓	✓				✓	✓	✓					
UI	✓	✓			✓	✓				✓	✓	✓					
Player	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓					
Enemy	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓					
CompanionAI	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓					
Trippers	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓					
StopMove	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓					
HeadTrack	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓					
BodyPart	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓					
BlockAIRayCast	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓					
Pushable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
CoverPoint	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓					
Vehicle	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓					
PostProcess	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓					

Cloth Inter-Collision ☐

15. WIP Features

16. Terrain Frictions

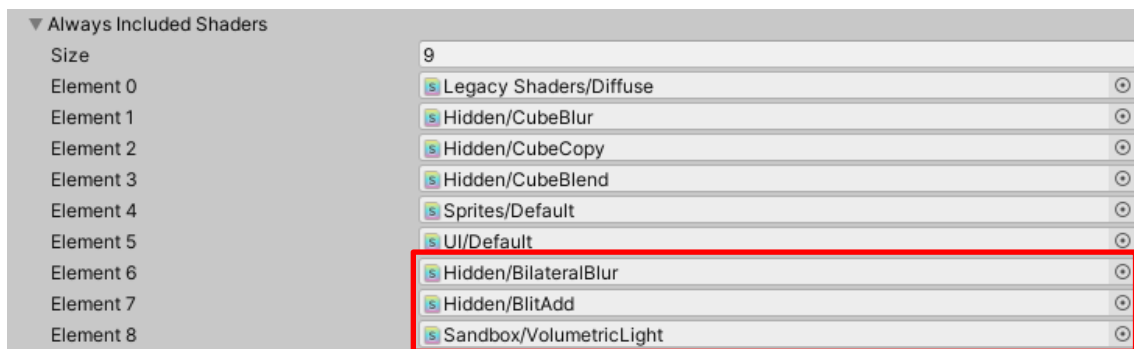
Various effects such as skidmarks and smoke are applied differently depending on the terrain splat map.

17. 2-Combo Unarmed WeakAttack

It is very difficult to attack enemies with hands while riding a motorcycle. This feature will be applied in consideration of the future situation.

18. Build

When building with Volumetric Headlight, the shaders must be set as follows:



<Project Settings - Graphics>

19. [MIS YouTube](#)

4. Upgrade

Please refer to Release History section in store page whenever this package gets upgrade.

The upgrade proceeds when any of the following occurs:

- Invector/MIS upgrade
- Bug fix or feature upgrade itself
- Compatibility with other MIS Package releases

IMPORTANT: Should check that packages with dependencies are released.

IMPORTANT: Check the Asset Store's Release History whether or not you should remove the old version.

IMPORTANT: Keep the install sequence with dependencies.

5. Contacts Information

Please contact us through the following channels:

- Email: mymobilin@gmail.com
- Discord: <https://discord.gg/CXrbdUNetc> (Recommended)

IMPORTANT: INVECTOR Company is not responsible for management for this asset



Make Game Better